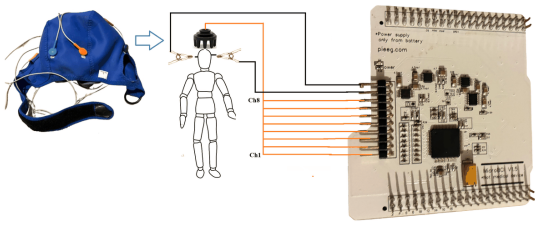
CONNECTOR PINOUT & ELECTRODE GUIDE · SKU PIA24024P

# Electrodes, Header & Nucleo Pins

How to connect 8 electrodes, the reference and the bias to MicroBCI, the default 10-20 montage, the STM32WB55 pins the shield uses on the ST morpho headers, and the LEDs and test points.



| 8 | REF | BIAS | 5 | 2.54 |
|---|---|---|---|---|
| Electrode channels | Ear clip · common 0 µV | Ear clip · noise rejection | MCU signal pins | mm header pitch |

## 01 ELECTRODE INPUTS

| Input | Count | Electrode | Purpose |
|---|---|---|---|
| **Channels 1–8** | 8 | Dry or wet Ag/AgCl, in the cap | Voltage between each electrode and the reference |
| **Reference** | 1 | Ear clip on one earlobe | Common 0 µV point for all 8 channels |
| **Bias** | 1 | Ear clip on the other earlobe | Drives the body to cancel common-mode noise such as 50/60 Hz hum |

## 02 ELECTRODE HEADER J3

Square pad = pin 1 (E1). 2.54 mm pitch, single row.

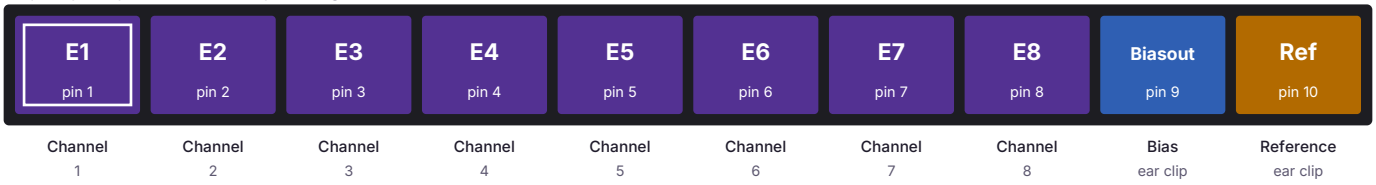



Figure 1 — Electrode header pin order. On the board E1 is nearest the bottom edge and Ref at the top.

**NOTE Reference and bias clips matter most.**

A loose ear clip makes every channel noisy at once. Clean the earlobes and make sure both clips sit firmly. For EMG or ECG, put reference and bias on a bony spot nearby (wrist, elbow).

## 03 LEDS AND TEST POINTS

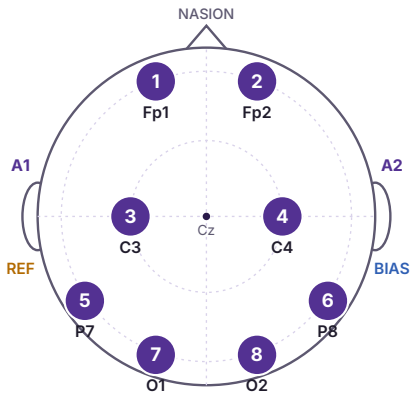
| LED | Meaning |
|---|---|
| **D4** "Power" | Shield powered |
| **D3** | Status (next to the ADS1299) |

| Point | Expected | Use |
|---|---|---|
| TP1 | ≈ −2.5 V | Supply check |
| TP2 | ≈ 3.3 V | Supply check |
| TP3 | ≈ +2.5 V | Supply check |

Measure against GND (CN7 pin 8) with a multimeter only, with no electrodes on anyone. Do not draw current from the test points.


For research and education only. Not a medical device. Specifications subject to change.

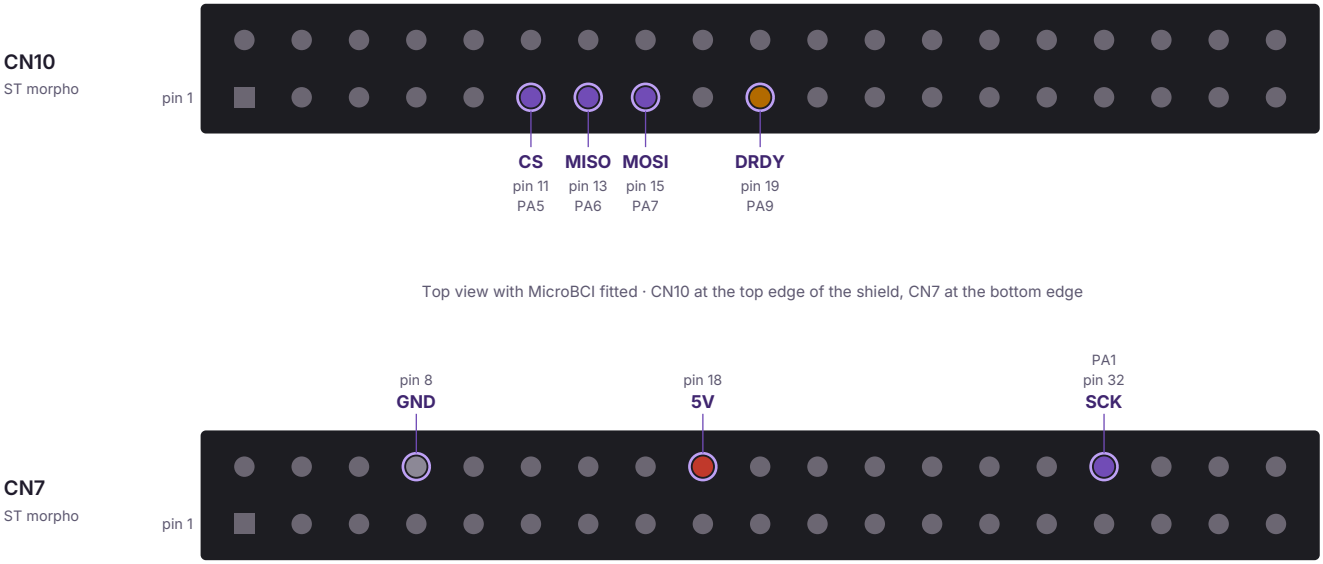
## 04 DEFAULT MONTAGE (10-20 SYSTEM)




| Ch | Position | Area | Good for |
|---|---|---|---|
| 1 | Fp1 | Front left | Blinks, eye movement |
| 2 | Fp2 | Front right | Blinks, eye movement |
| 3 | C3 | Centre left | Motor activity (right hand) |
| 4 | C4 | Centre right | Motor activity (left hand) |
| 5 | P7 | Back left, side | Parietal / temporal |
| 6 | P8 | Back right, side | Parietal / temporal |
| 7 | O1 | Back left | Alpha rhythm, vision |
| 8 | O2 | Back right | Alpha rhythm, vision |

The PiEEG 8-channel default. Any other positions work — note which channel went where.

## 05 NUCLEO-WB55 PINS (ST MORPHO)




Top view with MicroBCI fitted · CN10 at the top edge of the shield, CN7 at the bottom edge

| Morpho pin | STM32WB55 | Signal | ARDUINO® equivalent | Direction |
|---|---|---|---|---|
| CN7-32 | PA1 | SPI1 SCK | A2 | MCU → ADS1299 |
| CN10-13 | PA6 | SPI1 MISO (DOUT) | D12 | ADS1299 → MCU |
| CN10-15 | PA7 | SPI1 MOSI (DIN) | D11 | MCU → ADS1299 |
| CN10-11 | PA5 | Chip select (GPIO) | D13 | MCU → ADS1299 |
| CN10-19 | PA9 | DRDY — data ready, active low | D9 | ADS1299 → MCU |
| CN7-18 / CN7-8 | — | 5 V / GND | 5V / GND | Power to the shield |

Pin names from the supplied firmware and the NUCLEO-WB55 user manual (morpho table). If you write your own firmware, keep PA1, PA5, PA6, PA7 and PA9 free for the shield; CN10-15 and CN7-32 must stay on their default solder-bridge settings (PA7, PA1).

## 06 RELATED DOCUMENTS

| | | |
|---|---|---|
| **MB-DS-001**<br>Product Datasheet | **MB-MD-002**<br>Mechanical & Dimensional Drawing | **MB-PCB-003**<br>PCB & Manufacturing Specification |
| **MB-QSG-005**<br>Quick Start & Safety Guide | **MB-SW-006**<br>Firmware, Bluetooth & Software Guide | |


For research and education only. Not a medical device. Specifications subject to change.